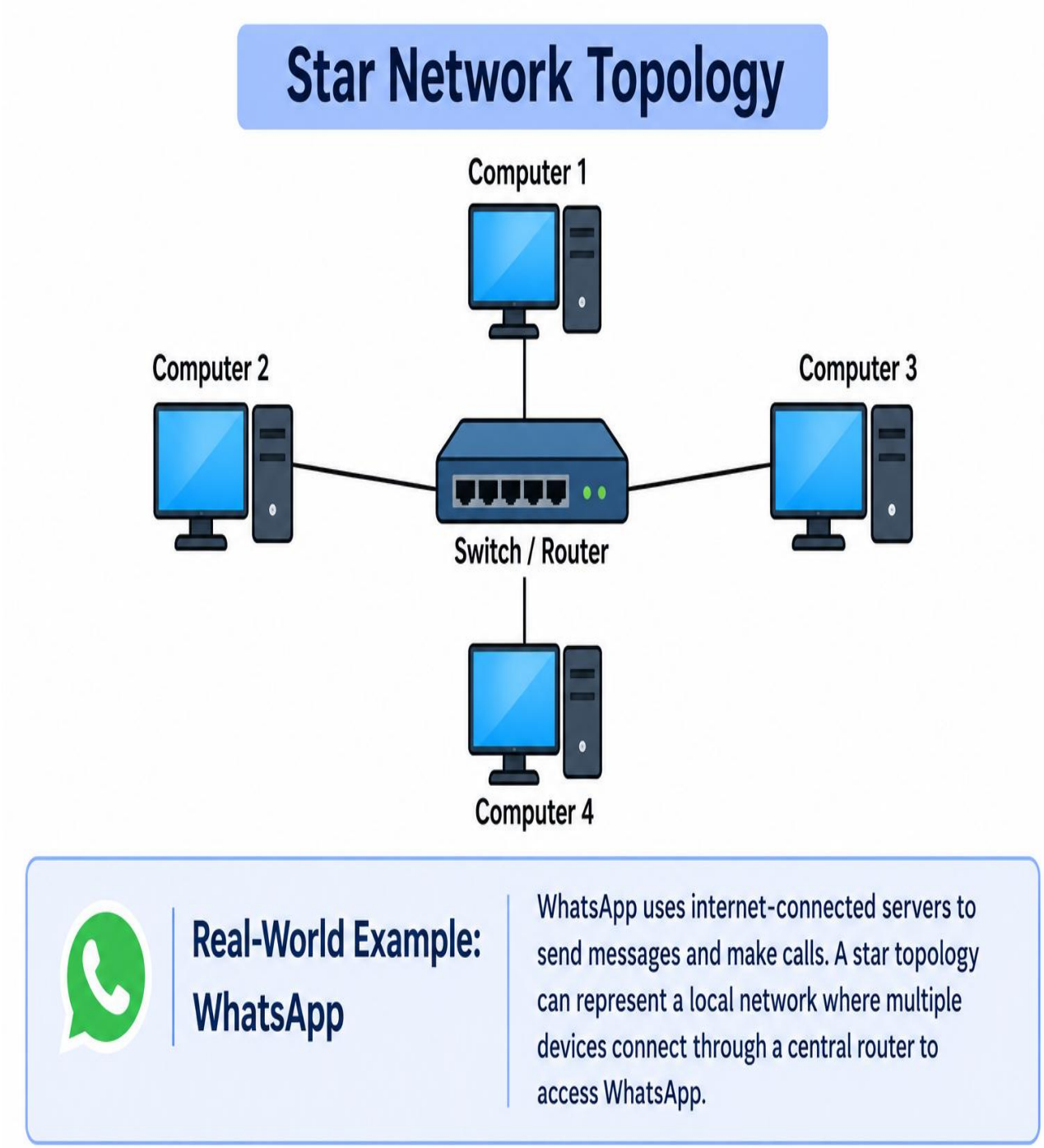
Network topologies

Q-1 Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

Ans : Network Topologiespicture

1. Star Network Topology

Diagram: Draw a central switch or router connected to four computers. Label the switch/router and all computers.

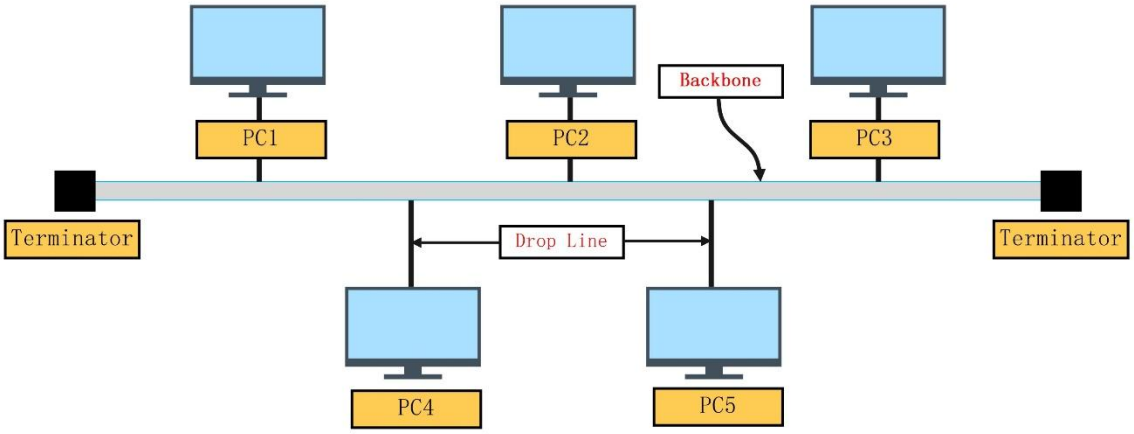


Real-World Example: WhatsApp

WhatsApp uses internet-connected servers to send messages and make calls. A star topology can represent a local network where multiple devices connect through a central router to access WhatsApp.

2. Bus Network Topology

Diagram: Draw one straight backbone cable and connect four computers to it. Label the computers and the backbone cable.

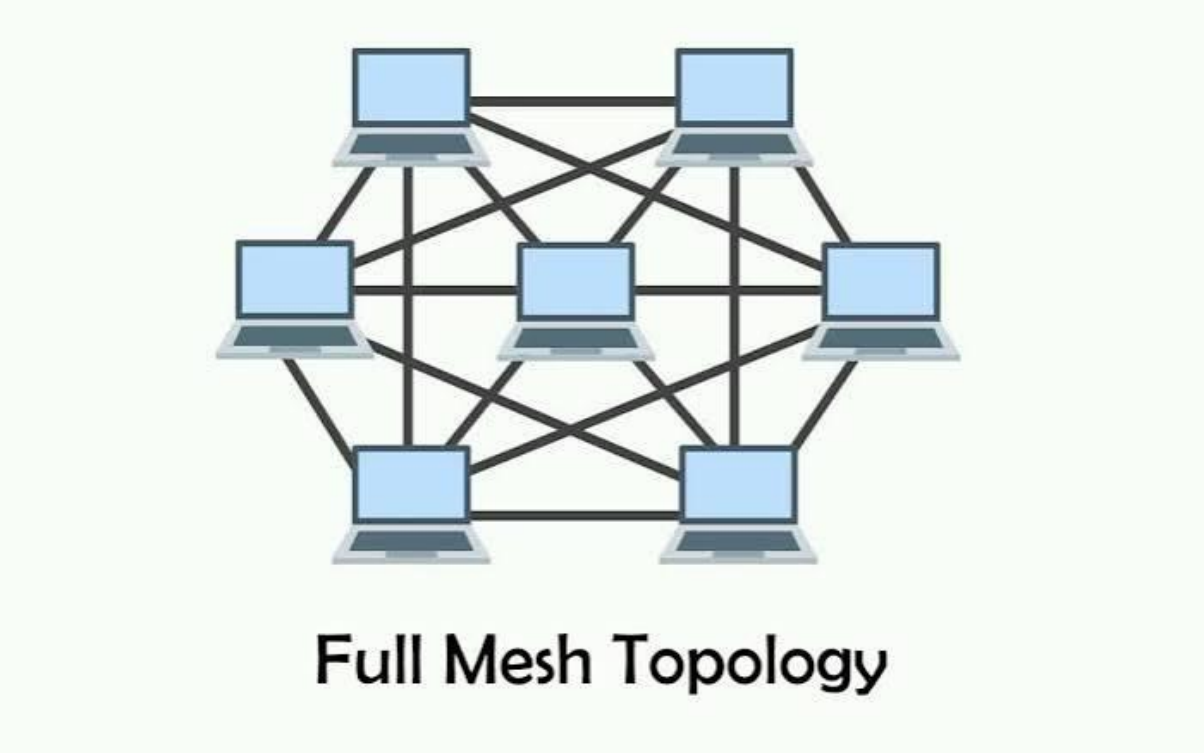


Real-World Example: Small Office LAN

A small office network can be represented using a bus topology, where multiple computers share one main cable. It is simple to set up, but if the main cable fails, the network may stop working.

3. Mesh Network Topology

Diagram: Draw six nodes and connect them with multiple links. Label each node and the connecting links.



Real-World Example: IPL Live Streaming

Mesh networking allows data to travel through alternative paths if one connection fails. This can improve network reliability, although IPL streaming services commonly use content delivery networks (CDNs) and internet routing rather than a pure mesh topology.

Q-2 Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Ans: Comparison of Network Topologies

Topology	Pros (Advantages)	Cons (Disadvantages)	Best Use Case with App Example
Star Topology	1. Easy to install and manage. 2. Easy to add new devices. 3. Failure of one computer does not affect others. 4. Easy to find and fix faults.	1. Requires more cables. 2. Needs a central switch or router. 3. If the central device fails, the network stops working. 4. Installation can be expensive for large networks.	Best use: Offices, colleges, and homes. Example: Instagram can be accessed by multiple devices connected to a central Wi-Fi router.
Bus Topology	1. Uses less cable. 2. Low installation cost. 3. Simple for small networks. 4. Does not require a central switch.	1. Failure of the backbone cable can stop the network. 2. Difficult to find faults in large networks. 3. Performance decreases when many devices are connected. 4. Limited cable length and number of devices.	Best use: Small or temporary networks. Example: A basic computer lab network can be used to demonstrate bus topology.
Ring Topology	1. Each device has an organized connection. 2. Data travels in a defined direction in a basic ring. 3. Can provide predictable performance under controlled traffic. 4. No central switch is required in a basic ring.	1. A single link or device failure can disrupt a basic ring. 2. Adding or removing devices may interrupt the network. 3. Troubleshooting can be difficult. 4. Delay may increase as data passes through more devices.	Best use: Specialized networks that need predictable data transmission. Example: A ring network can be used in a fiber-optic communication system.

Mesh Topology	1. Highly reliable because multiple paths are available. 2. Failure of one link may not stop communication. 3. Supports alternative routes for data. 4. Suitable for networks that require high availability.	1. Requires many cables or wireless links. 2. Expensive to install and maintain. 3. Complex to configure. 4. Requires more network hardware and management.	Best use: Reliable wireless or communication networks. Example: Paytm services can benefit from reliable network infrastructure, although Paytm itself does not depend on one specific topology.

Q-3 Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

Ans: **Suitable Network Topology for a Multiplayer Mobile Game**

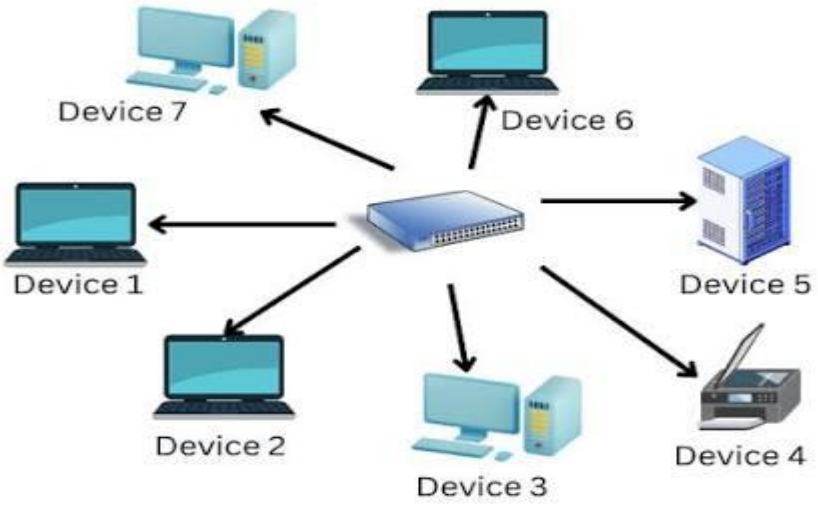
Chosen Topology: Star Topology

I would choose Star Topology for a multiplayer mobile game like PUBG or Free Fire because all game servers and devices can connect through a central network system. It provides fast communication and makes the network easier to manage. It can be reliable because a single device or connection failure may not affect the other devices, although failure of the central device can disrupt the network. The cost is moderate because it requires central networking equipment, but it is easier to maintain and expand.

Q-4 Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.

Hint: Consider factors like cost, ease of troubleshooting, and future expansion.

Ans:


Device 7
Device 6
Device 1
Device 5
Device 2
Device 3
Device 4